

1. SELECT count(distinct CompanyNumber)

FROM PayGap

	count(distinct CompanyNumber)
1	13378

2. SELECT count(CompanyNumber)
FROM PayGap
where submittedafterthedeathline is 'True'

	count(CompanyNumber)
1	4830

3. SELECT count(distinct CompanyNumber)
FROM PayGap
where companylinktogpginfo is null

	count(distinct CompanyNumber)
1	8942

4. SELECT
COUNT(*) AS total_rows,
SUM(CASE WHEN DiffMeanHourlyPercent IS NULL THEN 1 ELSE 0 END) AS
missing_DiffMeanHourlyPercent,
SUM(CASE WHEN DiffMedianHourlyPercent IS NULL THEN 1 ELSE 0 END) AS
missing_DiffMedianHourlyPercent,
SUM(CASE WHEN DiffMeanBonusPercent IS NULL THEN 1 ELSE 0 END) AS
missing_DiffMeanBonusPercent,
SUM(CASE WHEN DiffMedianBonusPercent IS NULL THEN 1 ELSE 0 END) AS
missing_DiffMedianBonusPercent,
SUM(CASE WHEN MaleBonusPercent IS NULL THEN 1 ELSE 0 END) AS
missing_MaleBonusPercent,
SUM(CASE WHEN FemaleBonusPercent IS NULL THEN 1 ELSE 0 END) AS
missing_FemaleBonusPercent,
SUM(CASE WHEN MaleLowerQuartile IS NULL THEN 1 ELSE 0 END) AS
missing_MaleLowerQuartile,

```

SUM(CASE WHEN FemaleLowerQuartile IS NULL THEN 1 ELSE 0 END) AS
missing_FemaleLowerQuartile,
SUM(CASE WHEN MaleLowerMiddleQuartile IS NULL THEN 1 ELSE 0 END) AS
missing_MaleLowerMiddleQuartile,
SUM(CASE WHEN FemaleLowerMiddleQuartile IS NULL THEN 1 ELSE 0 END) AS
missing_FemaleLowerMiddleQuartile,
SUM(CASE WHEN MaleUpperMiddleQuartile IS NULL THEN 1 ELSE 0 END) AS
missing_MaleUpperMiddleQuartile,
SUM(CASE WHEN FemaleUpperMiddleQuartile IS NULL THEN 1 ELSE 0 END) AS
missing_FemaleUpperMiddleQuartile,
SUM(CASE WHEN MaleTopQuartile IS NULL THEN 1 ELSE 0 END) AS
missing_MaleTopQuartile,
SUM(CASE WHEN FemaleTopQuartile IS NULL THEN 1 ELSE 0 END) AS
missing_FemaleTopQuartile
FROM PayGap

```

1	total_rows	missing_DiffMeanHourlyPercent	missing_DiffMedianHourlyPercent	missing_DiffMeanBonusPercent	missing_DiffMedianBonusPercent	missing_MaleBonu
0	83198	0	0	17237	17239	0
	missing_FemaleBonusPercent	missing_MaleLowerQuartile	missing_FemaleLowerQuartile	missing_MaleLowerMiddleQuartile	missing_FemaleLowerMiddleQuartile	
0		399	399	399	399	
	missing_MaleUpperMiddleQuartile	missing_FemaleUpperMiddleQuartile	missing_MaleTopQuartile	missing_FemaleTopQuartile		
	399	399	399	399		

5. select avg(DiffMeanHourlyPercent),avg(DiffMedianHourlyPercent)
from PayGap

	avg(DiffMeanHourlyPercent)	avg(DiffMedianHourlyPercent)
1	13.4126895947095	11.9255160898154

6. The median is the best way to calculate average salary because it ignores extreme highs and lows, giving a more accurate picture of what most people actually earn.

7. select count(*)
from PayGap
where DiffMedianHourlyPercent<0

count(*)
1 83199

Note: (from 8 to 11 same answer)

8. Select distinct CompanyNumber, DiffMedianHourlyPercent, CompnyLinkTo
GPGInfo, avg(MaleBounsPercent) as avg_male_bouns, avg(FemaleBounsPercent) as
avg_female_bouns
From PayGap
Where DiffMedianHourlyPercent is not Null
Group by 1
Having avg_male_bouns
Order by DiffMedianHourlyPercent desc
Limit 10

DB Browser for SQLite - C:\Users\DELL\OneDrive\Desktop\GA\SQL\Assignments\GenderGap\gender_gap_sq.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database

Database Structure Browse Data Execute SQL

SQL 1* SQL 2* SQL 3* SQL 4

```
1 select
2 DISTINCT CompanyNumber,
3 DiffMedianHourlyPercent, CompanyLinkToGPGInfo, avg(MaleBonusPercent) as avg_male_bouns, avg(FemaleBonusPercent) as avg_female_b
4 FROM G_g
5 WHERE DiffMedianHourlyPercent IS NOT NULL
6 group by 1
7 HAVING avg_male_bouns > avg_female_bouns
8 ORDER BY DiffMedianHourlyPercent DESC
9
10 LIMIT 10;
```

	CompanyNumber	DiffMedianHourlyPercent	CompanyLinkToGPGInfo	avg_male_bouns	avg_female_bouns
1	04761374	100.0	NULL	100.0	0.0
2	02207082	100.0	https://www.andersonsgroup.co.uk/app/...	12.0675	0.0
3	00418758	100.0	NULL	100.0	0.0
4	03561975	82.0	https://www.tmf-group.com/~media/...	14.15	10.45
5	00460129	78.4	NULL	54.18	43.92
6	07034946	73.0	NULL	2.7	0.4
7	07186841	72.0	NULL	77.98	25.965
8	04046987	71.0	NULL	90.5714285714286	76.7142857142857
9	06563486	69.0	NULL	10.375	6.1625
10	01617719	68.5	https://www.hgconstruction.co.uk/wp-...	26.27	16.92

DB Schema

Schema

- "PostCode" TEXT
- "CompanyNumber" TEXT
- "SicCodes" TEXT
- "DiffMean" "SicCodes" TEXT REAL
- "DiffMedianHourlyPercent" REAL
- "DiffMeanBonusPercent" REAL
- "DiffMedianBonusPercent" REAL
- "MaleBonusPercent" REAL
- "FemaleBonusPercent" REAL
- "MaleLowerQuartile" REAL
- "FemaleLowerQuartile" REAL
- "MaleLowerMiddleQuartile" REAL
- "FemaleLowerMiddleQuartile" REAL
- "MaleUpperMiddleQuartile" REAL
- "FemaleUpperMiddleQuartile" REAL
- "MaleTopQuartile" REAL
- "FemaleTopQuartile" REAL
- "CompanyLinkToGPGInfo" TEXT
- "ResponsiblePerson" TEXT
- "EmployerSize" TEXT
- "CurrentName" TEXT
- "SubmittedAfterTheDeadline" TEXT
- "DueDate" TEXT
- "DateSubmitted" TEXT

SQL Log DB Schema UTF-8

Type here to search

1:25 PM 8/30/2025

9. Select distinct CompanyNumber, DiffMedianHourlyPercent, CompnyLinkTo
GPGInfo, avg(MaleBounsPercent) as avg_male_bouns, avg(MaleBounsPercent) as
avg_female_bouns
From PayGap
Where DiffMedianHourlyPercent is not Null
Group by 1
Having avg_male_bouns
Order by DiffMedianHourlyPercent desc
Limit 10

The screenshot shows the DB Browser for SQLite application. The SQL query is as follows:

```

1 select
2 DISTINCT CompanyNumber,
3 DiffMedianHourlyPercent , CompanyLinkToGPGInfo ,avg(MaleBonusPercent) as avg_male_bouns , avg(FemaleBonusPercent) as avg_female_b
4 FROM G_g
5 WHERE DiffMedianHourlyPercent IS NOT NULL
6 group by 1
7 HAVING avg_male_bouns > avg_female_bouns
8 ORDER BY DiffMedianHourlyPercent DESC
9 LIMIT 10;

```

The results table displays the following data:

	CompanyNumber	DiffMedianHourlyPercent	CompanyLinkToGPGInfo	avg_male_bouns	avg_female_bouns
1	04761374	100.0	NULL	100.0	0.0
2	02207082	100.0	https://www.andersongroup.co.uk/app/...	12.0675	0.0
3	00418758	100.0	NULL	100.0	0.0
4	03561975	82.0	https://www.tmf-group.com/~media/...	14.15	10.45
5	00460129	78.4	NULL	54.18	43.92
6	07034946	73.0	NULL	2.7	0.4
7	07186841	72.0	NULL	77.98	25.965
8	04046987	71.0	NULL	90.5714285714286	76.7142857142857
9	06563486	69.0	NULL	10.375	6.1625
10	01617719	68.5	https://www.hgconstruction.co.uk/wp-...	26.27	16.92

The right-hand pane shows the DB Schema for the 'Schema' database, listing various tables and their columns, including 'PostCode', 'CompanyNumber', 'SicCodes', 'DiffMean', 'DiffMedianHourlyPercent', 'DiffMeanBonusPercent', 'MaleBonusPercent', 'FemaleBonusPercent', 'MaleLowerQuartile', 'FemaleLowerQuartile', 'MaleLowerMiddleQuartile', 'FemaleLowerMiddleQuartile', 'MaleUpperMiddleQuartile', 'FemaleUpperMiddleQuartile', 'MaleTopQuartile', 'FemaleTopQuartile', 'CompanyLinkToGPGInfo', 'ResponsiblePerson', 'EmployerSize', 'CurrentName', 'SubmittedAfterTheDeadline', 'DueDate', and 'DateSubmitted'.

10. Select distinct CompanyNumber, DiffMedianHourlyPercent, CompnyLinkTo
GPGInfo, avg(MaleBounsPercent) as avg_male_bouns, avg(MaleBounsPercent) as
avg_female_bouns
From PayGap
Where DiffMedianHourlyPercent is not Null
Group by 1
Having avg_male_bouns
Order by DiffMedianHourlyPercent desc
Limit 10

```

1 select
2   DISTINCT CompanyNumber,
3   DiffMedianHourlyPercent , CompanyLinkToGPGInfo ,avg(MaleBonusPercent) as avg_male_bouns , avg(FemaleBonusPercent) as avg_female_b
4   FROM G_g
5   WHERE DiffMedianHourlyPercent IS NOT NULL
6   group by 1
7   HAVING avg_male_bouns > avg_female_bouns
8   ORDER BY DiffMedianHourlyPercent DESC
9   LIMIT 10;

```

	CompanyNumber	DiffMedianHourlyPercent	CompanyLinkToGPGInfo	avg_male_bouns	avg_female_bouns
1	04761374	100.0	NULL	100.0	0.0
2	02207082	100.0	https://www.andersongroup.co.uk/app/...	12.0675	0.0
3	00418758	100.0	NULL	100.0	0.0
4	03561975	82.0	https://www.tmf-group.com/-/media/...	14.15	10.45
5	00460129	78.4	NULL	54.18	43.92
6	07034946	73.0	NULL	2.7	0.4
7	07186841	72.0	NULL	77.98	25.965
8	04046987	71.0	NULL	90.5714285714286	76.7142857142857
9	06563486	69.0	NULL	10.375	6.1625
10	01617719	68.5	https://www.hgconstruction.co.uk/wp-...	26.27	16.92

11. Select distinct CompanyNumber, DiffMedianHourlyPercent, CompnyLinkTo
 GPGInfo, avg(MaleBounsPercent) as avg_male_bouns, avg(MaleBounsPercent) as
 avg_female_bouns
 From PayGap
 Where DiffMedianHourlyPercent is not Null
 Group by 1
 Having avg_male_bouns
 Order by DiffMedianHourlyPercent desc
 Limit 10

```

1 select
2 DISTINCT CompanyNumber,
3 DiffMedianHourlyPercent , CompanyLinkToGPGInfo ,avg(MaleBonusPercent) as avg_male_bouns , avg(FemaleBonusPercent) as avg_female_b
4 FROM G_g
5 WHERE DiffMedianHourlyPercent IS NOT NULL
6 group by 1
7 HAVING avg_male_bouns > avg_female_bouns
8 ORDER BY DiffMedianHourlyPercent DESC
9 LIMIT 10;

```

	CompanyNumber	DiffMedianHourlyPercent	CompanyLinkToGPGInfo	avg_male_bouns	avg_female_bouns
1	04761374	100.0	NULL	100.0	0.0
2	02207082	100.0	https://www.andersongroup.co.uk/app/...	12.0675	0.0
3	00418758	100.0	NULL	100.0	0.0
4	03561975	82.0	https://www.tmf-group.com/-/media/...	14.15	10.45
5	00460129	78.4	NULL	54.18	43.92
6	07034946	73.0	NULL	2.7	0.4
7	07186841	72.0	NULL	77.98	25.965
8	04046987	71.0	NULL	90.5714285714286	76.7142857142857
9	06563486	69.0	NULL	10.375	6.1625
10	01617719	68.5	https://www.hgconstruction.co.uk/wp-...	26.27	16.92

12. WITH t1 AS (
 SELECT
 CompanyNumber,

```

DiffMedianHourlyPercent,
CASE
WHEN address LIKE '%london%' THEN 'London'
ELSE 'outside london'
END AS simple_address
FROM G_g
WHERE simple_address IS NOT NULL AND CompanyNumber is not NULL
)
SELECT AVG(DiffMedianHourlyPercent) AS avg_gap , simple_address
FROM t1
WHERE simple_address in('London','outside london')
group by 2

```

	avg_gap	simple_address
1	13.7441899160295	London
2	11.3361899245611	outside london

```

13. WITH t1 AS (SELECT
CompanyNumber,
DiffMedianHourlyPercent,
CASE
WHEN address LIKE '%birming%' THEN 'Birmingham'
WHEN address LIKE '%london%' THEN 'London'
ELSE NULL
END AS simple_address
FROM PayGap
WHERE simple_address IS NOT NULL and companynumber is not null
)
SELECT AVG(DiffMedianHourlyPercent) AS avg_gap , simple_address
FROM t1
WHERE simple_address in('London','Birmingham')

```

group by 2

	avg_gap	simple_address
1	10.1929485827664	Birmingham
2	13.7441899160295	London

Note: The bonus Q has been answered in Q14 and Q15.

Bonus:

```
14. select avg(G_g.DiffMedianHourlyPercent)
    from siccode inner join G_g on siccode."SIC Code" = G_g.SicCodes
    where siccode.Description like '%education%'
```

	avg(G_g.DiffMedianHourlyPercent)
1	21.8315271733132

Bonus:

```
15. select avg(G_g.DiffMedianHourlyPercent)
    from siccode inner join G_g on siccode."SIC Code" = G_g.SicCodes
    where siccode.Description like '%bank%'
```

	avg(G_g.DiffMedianHourlyPercent)
1	30.28209375

```
16. SELECT
    EmployerSize,
    CASE
    WHEN EmployerSize LIKE '%to%' THEN
    (
    CAST(substr(EmployerSize, 1, instr(EmployerSize, ' ') - 1) AS INTEGER) +
    CAST(substr(EmployerSize, instr(EmployerSize, 'to') + 3) AS INTEGER)
    ) / 2
    END AS EmployerSizeMid
    FROM PayGap
```

	EmployerSize	EmployerSizeMid
1	250 to 499	374
2	Less than 250	<i>NULL</i>
3	Less than 250	<i>NULL</i>
4	250 to 499	374
5	500 to 999	749
6	250 to 499	374
7	250 to 499	374
8	250 to 499	374
9	500 to 999	749
10	500 to 999	749
11	1000 to 4999	2999
12	250 to 499	374
13	1000 to 4999	2999
14	1000 to 4999	2999
15	250 to 499	374

Continuing Q16

WITH SizeMidpoints AS (

SELECT

EmployerSize,

DiffMedianHourlyPercent,

CASE

WHEN EmployerSize LIKE '%to%' THEN (

CAST(SUBSTR(EmployerSize, 1, INSTR(EmployerSize, ' ') - 1) AS INTEGER) +

CAST(SUBSTR(EmployerSize, INSTR(EmployerSize, 'to') + 3) AS INTEGER)

) / 2

WHEN EmployerSize LIKE 'Less than %' THEN

CAST(SUBSTR(EmployerSize, LENGTH('Less than ') + 1) AS INTEGER)

ELSE null

END AS EmployerSizeMid

FROM G_g

WHERE DiffMedianHourlyPercent IS NOT NULL

),

Grouped AS (

SELECT

CASE

WHEN EmployerSizeMid <= 500 THEN '0–500'

WHEN EmployerSizeMid <= 1000 THEN '501–1000'

WHEN EmployerSizeMid > 1000 THEN '1001+'

ELSE 'Unknown'

END AS SizeGroup,

DiffMedianHourlyPercent

FROM SizeMidpoints

WHERE EmployerSizeMid IS NOT NULL

)

SELECT

SizeGroup,

AVG(DiffMedianHourlyPercent) AS AvgGap

FROM Grouped

GROUP BY SizeGroup

ORDER BY

CASE

```
WHEN SizeGroup = '0-500' THEN 1
WHEN SizeGroup = '501-1000' THEN 2
WHEN SizeGroup = '1001+' THEN 3
ELSE 4
END;
```

SizeGroup	AvgGap
0-500	12.244691379971
501-1000	12.3628200605717
1001+	10.9433624545645

That's mean that there is a small relationship between the number of employess and the avg gap so this is not the reason.

Presentation codes

--Intro

```
SELECT count(distinct CompanyNumber)
FROM PayGap
```

```
SELECT
sum (CASE
WHEN DiffMedianHourlyPercent < 0 then 1 else 0 end) as girl ,
sum (CASE
WHEN DiffMedianHourlyPercent = 0 then 1 else 0 end) as equality ,
sum (CASE
WHEN DiffMedianHourlyPercent > 0 then 1 else 0 end) as boys

from G_g
```

--Q1 avg across UK

```
SELECT
round(AVG(FemaleLowerQuartile)) AS avg_female_low,
round(AVG(FemaleLowerMiddleQuartile)) AS avg_female_lower_mid,
round(AVG(FemaleUpperMiddleQuartile)) AS avg_female_upper_mid,
round(AVG(FemaleTopQuartile)) AS avg_female_top,
round(AVG(maleLowerQuartile)) AS avg_male_low,
round(AVG(maleLowerMiddleQuartile)) AS avg_male_lower_mid,
round(AVG(maleUpperMiddleQuartile)) AS avg_male_upper_mid,
round(AVG(maleTopQuartile)) AS avg_male_top
FROM G_g;
```

```
select avg(DiffMedianHourlyPercent) as difference_median_avg
from G_g
```

--Q2 avg pay gap industries

```
WITH t1 AS (  
    SELECT AVG(G_g.DiffMedianHourlyPercent) AS avg_bank  
    FROM siccode  
    INNER JOIN G_g ON siccode."SIC Code" = G_g.SicCodes  
    WHERE siccode.Description LIKE '%bank%'  
),  
t2 AS (  
    SELECT AVG(G_g.DiffMedianHourlyPercent) AS avg_school  
    FROM siccode  
    INNER JOIN G_g ON siccode."SIC Code" = G_g.SicCodes  
    WHERE siccode.Description LIKE '%education%'  
)  
,  
t3 AS (  
    SELECT AVG(G_g.DiffMedianHourlyPercent) AS avg_resturants  
    FROM siccode  
    INNER JOIN G_g ON siccode."SIC Code" = G_g.SicCodes  
    WHERE siccode.Description LIKE '%restaurant%'  
)  
,  
t4 as (  
    SELECT AVG(G_g.DiffMedianHourlyPercent) AS avg_hospital  
    FROM siccode  
    INNER JOIN G_g ON siccode."SIC Code" = G_g.SicCodes  
    WHERE siccode.Description LIKE '%medical%'  
    )  
,  
    t5 as (SELECT avg(G_g.DiffMedianHourlyPercent) AS  
avg_manufaturing_industies  
    FROM siccode  
    INNER JOIN G_g ON siccode."SIC Code" = G_g.SicCodes  
    WHERE siccode.Description LIKE '%manufacture%')  
  
SELECT t1.avg_bank, t2.avg_school , t3.avg_resturants ,  
t4.avg_hospital,t5.avg_manufaturing_industies  
FROM t1, t2,t3,t4,t5;
```

```

--Q3 countries
WITH base AS (
SELECT
REPLACE(UPPER(TRIM(PostCode)), ' ', '') AS pc,
DiffMedianHourlyPercent AS diff
FROM G_g
WHERE DiffMedianHourlyPercent IS NOT NULL
),
tagged AS (
SELECT
diff,
CASE

WHEN SUBSTR(pc,1,2) IN ('NW','SE','SW','EC','WC') THEN 'London'
WHEN SUBSTR(pc,1,1) IN ('N','E','W') THEN 'London'


WHEN SUBSTR(pc,1,2) IN
('RG','OX','HP','SL','KT','TW','GU','RH','BN','ME','DA','TN','PO','SO','BR','CR','SM') THEN 'South
East'


WHEN SUBSTR(pc,1,2) IN ('BS','BA','EX','TR','PL','TA','DT','SP','SN','TQ') THEN 'South West'


WHEN SUBSTR(pc,1,2) IN ('CV','DY','WS','WV') THEN 'West Midlands'
WHEN SUBSTR(pc,1,1) = 'B' THEN 'West Midlands'


WHEN SUBSTR(pc,1,2) IN ('LE','NG','DE','LN','DN','NN','PE') THEN 'East Midlands'


WHEN SUBSTR(pc,1,2) IN ('CB','CM','CO','NR','IP','SG','AL','LU','SS') THEN 'East of
England'


WHEN SUBSTR(pc,1,2) IN ('PR','FY','BB','OL','BL','WA','WN','CW','CH') THEN 'North West'

```

WHEN SUBSTR(pc,1,1) IN ('L','M') THEN 'North West'

WHEN SUBSTR(pc,1,2) IN ('BD','LS','YO','HU','HX','WF','HD','DN') THEN 'Yorkshire and the Humber'

WHEN SUBSTR(pc,1,2) IN ('NE','SR','DH','DL','TS') THEN 'North East'

WHEN SUBSTR(pc,1,2) IN
('EH','AB','DD','DG','FK','G','HS','IV','KA','KW','KY','ML','PA','PH','ZE') THEN 'Scotland'
WHEN SUBSTR(pc,1,1) = 'G' THEN 'Scotland'

WHEN SUBSTR(pc,1,2) IN ('CF','NP','SA','LD','LL','SY') THEN 'Wales'

WHEN SUBSTR(pc,1,2) = 'BT' THEN 'Northern Ireland'

ELSE 'Other / Unknown'

END AS region

FROM base

)

SELECT

region,

AVG(diff) AS avg_DiffMedianHourlyPercent

FROM tagged

GROUP BY region

ORDER BY avg_DiffMedianHourlyPercent DESC;

	region	avg_DiffMedianHourlyPercent
1	South West	13.8450542777778
2	Yorkshire and the Humber	13.6154764827586
3	London	12.7528376555559
4	South East	12.4424146369688
5	North East	12.3252397498263
6	East of England	12.2590789249661
7	East Midlands	11.6631842888509
8	West Midlands	10.9296750922782
9	Other / Unknown	10.8790797964224
10	North West	9.98555964065289
11	Scotland	9.62310771793448
12	Wales	9.12128967828418

```

--year
WITH base AS (
SELECT
CASE
    WHEN DateSubmitted GLOB '????*'
    THEN CAST(SUBSTR(DateSubmitted, 1, 4) AS INTEGER)

    ELSE NULL
END AS Year,
DiffMedianHourlyPercent AS diff
FROM G_g
WHERE DiffMedianHourlyPercent IS NOT NULL
),
agg AS (
SELECT
Year,
AVG(diff) AS avg_DiffMedianHourlyPercent
FROM base
WHERE Year BETWEEN 2018 AND 2025
GROUP BY Year

```

```

)
SELECT
Year,
avg_DiffMedianHourlyPercent
FROM agg
ORDER BY Year;
--Q4
WITH t AS (
    SELECT CurrentName, AVG(DiffMedianHourlyPercent) AS diff
    FROM G_g
    WHERE DiffMedianHourlyPercent IS NOT NULL
    GROUP BY CurrentName
),
top10 AS (
    SELECT * FROM t
    ORDER BY diff DESC
    LIMIT 10
)
SELECT
    top10.CurrentName,
    top10.diff,
    siccode.Description
FROM top10
JOIN G_g ON G_g.CurrentName = top10.CurrentName
JOIN siccode ON siccode."SIC Code" = G_g.SicCodes;

```

Bonus slide:

```

SELECT round(avg(FemaleBonusPercent),2) as Female_Bouns,
round(avg(MaleBonusPercent),2) as Male_Bouns

```

from G_g

	Female_Bouns	Male_Bouns
1	36.06	37.14